

Automatic LED Greeting Card Project

Aim

To design a greeting card that automatically turns ON an LED when the card is opened and turns OFF the LED when the card is closed.

Problem Statement

Conventional greeting cards are not interactive and do not provide any visual effect when opened. A simple mechanism is needed to automatically light an LED when the card is opened and switch it off when the card is closed, making the card more attractive and engaging.

Proposed Solution

A paper-based switch is integrated into the greeting card using aluminum foil conductors. When the card is closed, a paper spacer breaks the circuit, keeping the LED OFF. When the card is opened, the foil contacts touch each other, completing the circuit and turning the LED ON automatically.

Components Required

- LED
- CR2032 Coin Cell (3V)
- Aluminum Foil
- A4 Sheet (folded as greeting card)
- Colour Paper
- Sketch Pens
- Tape or Glue
- Small Paper Strip (for switch mechanism)
- Circuit Working
- The CR2032 battery supplies power to the LED.
- Aluminum foil acts as the conductor.
- A paper switch is placed between two foil contacts.
- When the greeting card is closed, the paper strip separates the foil contacts, creating an open circuit and keeping the LED OFF.
- When the greeting card is opened, the foil contacts touch each other, completing the circuit and allowing current to flow through the LED.
- The LED lights up automatically when the card is opened.

Procedure

1. Fold the A4 sheet to form a greeting card.
2. Design and decorate the card using colour paper and sketches.
3. Create conductive paths using aluminum foil.
4. Connect the LED and coin cell battery to the foil circuit.
5. Make a simple paper switch near the fold of the card.
6. Position the switch so that the contacts touch when the card is opened.
7. Place a paper spacer so the contacts separate when the card is closed.
8. Test the card by opening and closing it.
9. Observe the LED turning ON when opened and OFF when closed.

Applications

- Interactive greeting cards
- Birthday cards
- Festival cards
- School science projects
- Creative electronic crafts

Advantages

- Automatic operation
- Low cost
- Easy to build
- No programming required
- Educational and creative
- Lightweight and portable

Result

The LED automatically glows when the greeting card is opened and turns OFF when the greeting card is closed.

Conclusion

The automatic LED greeting card successfully demonstrates the concept of a paper-based switch and basic electrical circuits. It enhances the appearance of greeting cards while providing a simple and practical application of electronics.

Project diagram

